

**| RESEARCH ARTICLE****Effects of *Sardinella maderensis* oil on inflammatory, hematological, oxidative stress parameters and histopathological changes in high fat diet obese-induced rats****Deffo Tiepma Ngongang Eurydice Flore | Ndifon Nahgwa Berkley | Achidi Aduni Ufuan | Arrey Oben Ebob | Tiencheu Bernard*****| Authors' Affiliation**

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| Abstract

Obesity results from excessive accumulation of fat in the body. Hence, this work aimed at evaluating the effect of sea *Sardinella maderensis* oil on oxidative stress, hematology, inflammatory and histopathology of obese rats. Oil was extracted from *Sardinella maderensis* and its quality indices assessed. The biological activity of the oil was investigated using 36 male albino rats, grouped into 6 rats per group. Group 1 was the normal group, Groups 2 to 6 received high fat diet to induce obesity. Also, group 3 received Orlistat®, while groups 4, 5 and 6 received fish oil at different concentrations for 28 days. At day 28, The rats were sacrificed, whole blood collected for serum preparation which was used for hematology and inflammatory cytokine evaluation. Organs were collected, weighed and used for oxidative stress and histopathological evaluation. Results revealed that values for oil quality indices were within the range of standard by Codex Alimentarius. Heart MDA levels in the negative control group (34.69µmol/g) were only significantly higher than test group III, 20.01µmol/g. Liver GSH levels in the negative control group (4.81µmol/g) were significantly lower than test group III, 8.35µmol/g. Catalase activity in the negative control group of the heart, 12.76U/mg, were significantly lower than in test groups I, II and III. Hematological parameters were all significantly lower in the negative control group than the test groups. Serum levels of pro-inflammatory cytokines in the negative control group, TNF-α (374.4), INF-δ (380.91) and IL-1β (334.45 pg/ml) were significantly higher than in test group II and III. Histopathological studies in the negative control group showed restoration of the defects in the test groups administered fish oil. Hence, *Sardinella maderensis* oil have a role in protection from the effects of obesity on oxidative stress, hematology, inflammatory cytokines and histology.

Keywords: Inflammatory Cytokine | Obesity | Oxidative Stress | *Sardinella maderensis* oil

1 | Introduction

Excessive accumulation of fat in the body results in obesity and overweight (Waxman, 2004), a rapidly growing public health problem (James, 2018). Obesity is mainly caused by poor diet management and physical inactivity. It is affecting an increasing number of countries worldwide because of its prevalence, costs and health effects (Alfariset al., 2023). In Cameroon, the burden of overweight and obesity is high, putting many at an increased risk of developing associated metabolic and cardiovascular complications (Nansseu et al., 2019). In 2005, 30.0% of the world's adult

population was obese, this value is estimated to almost double by 2030 (Simo et al., 2021; Kelly et al., 2008). The number and size of adipose tissues (AT) increases in cases of obesity (Nishimura et al., 2009), bringing about a state of chronic, low-grade inflammation (Ferrante, 2007; Gregor & Hotamisligil, 2011) which results in oxidative stress. An increase in oxidative stress-derived inflammation has been hypothesized to be a major mechanism in the pathogenesis and progression of obesity-related disorders (Elmarakby & Sullivan, 2010). When energy intake exceeds energy expenditure, metabolic



dysfunctions will be stimulated by responses triggered in many cell types such hepatocytes, adipocytes and immune cells (Hotamisligil, 2006; Lowell & Shulman, 2005). The generation of reactive oxygen species (ROS) in the body is an example of an adverse cellular response to excess nutrient (Bondia-Pons *et al.*, 2012). However, decreasing AT inflammation could be a therapeutic approach to managing obesity and its effects. Various dietary compounds including long chain omega-3 fatty acids (Calder, 2006), antioxidants (Del Rio *et al.*, 2011), prebiotics and probiotics (Latvala *et al.*, 2011; O'Hara *et al.*, 2006) have the ability to inflect predisposition to inflammatory conditions and may have a role in its management (Calder *et al.*, 2009). Omega-3 fatty acids for example eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) are reported to have chemo-protective and chemotherapeutic effects on diseases that involve inflammation (Friedman & Moe, 2006). EPA and DHA are from fish oil supplements, fatty fishes like mackerel and salmon (Nanounu *et al.*, 2011; Go *et al.*, 2016). In physiological conditions, cells exhibit a self-protective antioxidant activity against oxidative damage. These antioxidants could be enzymatic (superoxide dismutase, catalase, glutathione peroxidase) or non-enzymatic (vitamins E and C) (Valdecantos *et al.*, 2009). An increase in the generation of ROS results in an imbalance between the antioxidant and oxidant factors, favoring the latter. Thus, stimulating a pro-oxidative condition (Hopps *et al.*, 2010). The damage oxidative stress causes to cellular structures triggers an inflammatory response (Hotamisligil, 2006; Roberts & Sindhu 2009). It has been associated with adiposity, insulin resistance, and metabolic syndrome (Furukawa *et al.*, 2004; Meigs *et al.*, 2007), suggesting that oxidative stress could be an early event in the

pathology of these chronic diseases (Ceriello & Motz, 2004). Obesity is considered as an inflammatory disorder and omega-3 PUFAs have been proven to have anti-inflammatory effect (Rizos & Elisaf, 2017). Therefore, including dietary supplements with anti-inflammatory roles in the diets of obese individuals is seen as a promising approach to managing obesity (de Mello *et al.* 2018) and its effects on health. A negative effect was derived with the use of omega-6 fatty acids as they are generally pro inflammatory mediators. They promote fat accumulation and weight gain (Birch *et al.* 2002; Nuernberg *et al.* 2011). High intake of omega-3 fatty acids tends to alleviate the intensity of pro-inflammatory eicosanoids production, thereby preventing inflammation (Martínez-Fernández *et al.* 2015). Atlantic sardine, precisely *Sardinella maderensis* is commonly consumed in Cameroon and *Sardinella maderensis* have been reported to be rich in omega-3 PUFA. Hence, this work aimed at evaluating the effect of *Sardinella maderensis* fish oil in obese rats. In Cameroon people consume a lot of fish rich in omega-3 PUFAs, but the fish is not eaten raw and during cooking the polyunsaturated fatty acid (PUFA) may have been destroyed. Fish oil contains important omega-3 fatty acids proven to easily be absorbed. Additionally, it is still undetermined how fatty acid from sardine fish can prevent the obesity. Thus, the study verified the effect of treating obesity-induced rats with fish oil. Specifically to extract oil from *Sardinella maderensis* and assess its quality, followed by evaluating and comparing the inflammatory cytokine and oxidative stress and hematology of treated versus untreated rats. Lastly carrying out a histopathological study on treated versus untreated rats models.

2| Materials and methods

2.1| Sample Collection

Twenty-five kilograms of freshly captured Atlantic Sardine fish (*Sardinella maderensis*) was bought from CONGELCAM Buea, South West Region of Cameroon. They were placed in ice with a fish/ice ratio of 1:2 (w/w), transported to the Agro-Ecology Laboratory, University of Buea and kept in the freezer (-20°C) for future use.

2.2| Sample Preparation

Prior to its use, the fish was thawed using tap water, its scales and fins were removed. The fish was washed to remove residual blood. Fish fillet was obtained by cutting the fish lengthwise along the backbone to obtain maximum flesh.

2.3| Oil Extraction

Oil was extracted using the Bligh and Dyer method (AOAC, 1995). The cleaned fish was ground and mixed with water, methanol and chloroform in solvent ratios 1:2:1 respectively. The mixture was blended, squeezed using a clean linen cloth and filtered using a whatman filter paper number 4. The polar phase, water and methanol were removed using separating funnel. The chloroform solvent (lower liquid layer) was collected and filtered using a whatman filter paper grade 2.5. Oil was obtained by evaporating the solvent at 40°C using a vacuum rotatory evaporator under reduced pressure.

2.4| Oil Quality Assessment

The physicochemical properties of the oils were determined to assess its quality and purity. Acid value (free fatty acid) and Iodine value determination were carried out using the method described by A.O.A.C (1995). P-Anisidine value was determined in duplicate based on Tompkins

& Perkins (1999). Peroxide value determination was carried-out using the method described by Ranken (1988). The remaining tests such as iodine value, p-anisidine value and TOTOX value were carried out in duplicate using standard methods of AOAC (1995).

2.5| Animal Bioassay

2.5.1| Animal Purchase and ethical consideration

Male albino rats (150-200 g, 3-4 months old) used for this research were purchased at the animal house of the Department of Zoology and Animal Physiology, University of Douala, Cameroon. These animals were grouped and kept in the animal house and allowed to acclimatize (12:12 h light/dark cycle) for one week in metabolic cages (containing sawdust) with water and food (basalt diet) given *ad-libitum* before the commencement of the experiment. In order to carry out this study, an ethical clearance for animal handling and care was obtained from the University of Buea-Institutional Animal Care and Use Committee, UB- IACUC N° 08/2023.

2.5.2| Animal Grouping

Thirty-six rats were randomly divided into six groups of six animals each (Table 1). Group 1 (Normal Group) was fed with normal rodent chow. Groups 2 to 6 were fed high fat diet (HFD) (Table 2) for 56 days to induce obesity and fish oil was administered for 28 days following induction. The growth response of each rat was measured weekly during the 28 days. After 28 treatment days (at the end of the experiment), the rats were fasted overnight before being anesthetized by intraperitoneal administration with Diazepam (10 mg/kg) and ketamine (60 mg/kg), euthanatized and sacrificed.

Table 1: Animal groupings and Experimental design

Serial numbers		Groups	Treatments
Group 1		Normal Group	Healthy rats fed with normal rodent chow + distilled water
Group 2		Negative Control	Fed HFD + Distilled water 250mg/kg body weight
Group 3		Positive Control	Fed with HFD + Oral administration of 20 mg/kgBW Orlistat® for the last 28 days consecutive (Hassan <i>et al.</i> , 2021).
Group 4, 5 and 6		Test Groups	Fed with HFD + 250, 500, 1000mg/kg body weight of <i>Sardinella maderensis</i> oil respectively by oral gavage according to Waranya <i>et al.</i> , (2016) using a curved 18-gauge, 2.25 ball diameter and 2 inches in length feeding needle.

2.5.3| Animal Feed Preparation

Standard feed was composed according to Othman *et al.* (2019). Both HFD and normal rodent chow were prepared every after two days to prevent lipid oxidation. For every 1kg of animal feed, both HFD and normal rodent chow composition are on table 2.

2.5.4| Organ and Whole Blood Collection

Table 2: Standard animal feed preparation

Normal Rodent Chow			High Fat Diet		
Component	Amount (g)		Component	Amount (g)	
Maize flour	600		Normal rat chow	600	
Soya bean	200		Lard	290	
Fish powder	190		Boiled egg yolk	100	
Salt/Vitamins mix	10		Salt/Vitamins mix	10	

Anaesthetized animals were euthanized after the blood collection. Animals were placed in a recumbent supine position on the animal operation table. An incision was made on the ventral aspect of the thoracic wall just above the diaphragm by using a surgical scalpel, and the thoracic cavity was opened. The pancreas, heart, liver, brain, lungs, spleen and kidney were collected by dissecting it out from the major blood vessels. The organs were weighed and their relative organ weights were calculated by dividing

each animal's organ weight by their body weight. The organs for histological studies were stored in formalin and organs for oxidative stress evaluation were used to prepare organ homogenates.

2.5.5 | Biological and Biochemical Analysis

2.5.5.1 | Oxidative Stress

Its levels were evaluated using oxidative stress biomarkers both in serum and in organ homogenates.

Preparation of organ homogenate

Using the liver, heart, pancreas and brain, 22% organ homogenates were prepared. The organ samples were crushed in a mortar, using a pestle. Based on the mass of the organs, a particular volume of Tris-HCl buffer (50 mM, pH 7.4) was added to the crushed tissue. The mixture was centrifuged at 3000 rpm for 10mins and the supernatant was collected using a pipette. The organ homogenates of the various groups were used to evaluate levels of Malondialdehyde (MDA), Reduced Glutathione (GSH), Superoxide Dismutase (SOD), Catalase (CAT) and Nitrite as biomarkers of oxidative stress.

Evaluation of Malondialdehyde (MDA) levels

MDA was measured according to the method described by Wilbur *et al.* (1990). Briefly, 500 µL of organ homogenate (22%) and serum were added to a solution containing trichloroacetic acid (20%, 250 µL) and thiobarbituric acid (0.67%, 500 µL). The mixture was incubated under high temperature for 1 hour and then cooled under tap water. The mixture was then centrifuged at 3000 rpm for 10 minutes and the absorbance of the supernatant read at 530 nm. The concentration of MDA was quantified using an extinction coefficient of $1.56 \times 10^5 \text{ M}^{-1}\text{cm}^{-1}$ and expressed as µmol of MDA per g of tissue (Temdie *et al.*, 2020).

$$\text{MDA (mM)} = (A_{530} - A_{600}) / \epsilon$$

Where; A= Absorbance, ϵ = Extinction coefficient = $1.56 \times 10^5 \text{ M}^{-1}\text{cm}^{-1}$

Evaluation of Reduced Glutathione (GSH) activity

Reduced glutathione (GSH) level in organ homogenates and serum was assayed as described by Ellman (1959). Briefly, 20 µL of organ homogenates and serum were added to 3 mL of Ellman reagent. The mixture was kept at room temperature for 1 h, and its absorbance was read at 412 nm (Kouémou *et al.*, 2017). The concentration of GSH was quantified by the extinction coefficient (ϵ) of $13600 \text{ M}^{-1}\text{cm}^{-1}$ and expressed as µmol of tissue. The concentration of GSH was determined using the formula: $A = \epsilon CL$

Where; A = absorbance, L = path length in centimetres, C = concentration in Moles/litter (M), ϵ the extinction coefficient.

Evaluation of Superoxide dismutase (SOD) activity

The activity of SOD was evaluated according to the method described by Misra & Fridovich, (1972). 50.25 µL of organ homogenate, 625 µL buffer (0.05 M carbonate buffer, pH 10.2), and

75µL epinephrine (0.1 M HCl) were added to the spectrophotometric cuvette. The same was done for serum. Immediately after epinephrine was introduced into the medium, the absorbance was read at 480 nm after 20 and 80 s. Changes in absorbance were compared with controls where the sample was replaced with NaCl. The increase in absorbance was then calculated into unit of SOD activity (U), where one-unit of SOD activity is equivalent to the quantity of SOD that caused a 50% inhibition of the auto- oxidation of

epinephrine. The specific activity of SOD was expressed as SOD Unit/mg of tissue.

$$\text{SOD(U/mg)} = A_{\text{blank}} - A_{\text{control}} - (A_{\text{sample}} - A_{\text{sample control}}) * 100$$

Evaluation of levels of Catalase (CAT) in organ homogenate and serum

The activity of catalase was assayed according to the method of [Takahara *et al.* \(1960\)](#). 300 µL of phosphate buffer was placed in a cuvette, followed by the addition of 50 µL of the sample (homogenate (liver, heart, pancreas and brain) and serum). The reaction was started by the addition of 200 µL of H₂O₂. The absorbance was immediately read at 30 s intervals for 3mins at 240 nm upon addition of H₂O₂. For the blank, 300 µL of phosphate buffer, 50 µL of distilled water to replace the sample and 200 µL of distilled water to replace H₂O₂.

A molar absorptivity of 43.6 M cm⁻¹ was used to determine CAT activity. The activity of enzyme was expressed as µmoles of H₂O₂ decomposed/min/mg of tissue.

$$\text{Catalase activity} = (A * 10 * V_t) \times 100 / (1 * \epsilon * V_s)$$

Where A is the absorbance, V_t; total volume in the medium, V_s; volume of sample and ϵ ; extinction coefficient

Evaluation of levels of Nitrite in organ homogenate and serum

Nitrite content was determined in organ homogenate (22%) and in serum by Griess reagent (1% sulphanilamide and 0.1% naphthylethylenediamide in 2.5% phosphoric acid) ([Temdie *et al.*, 2012](#)). 100 µL of tissue homogenate each and also serum was mixed with 100 µL of Griess reagent. 100 µL of blank (50 mM, pH7.4 tris HCl). The mixture was kept at room temperature for 5 min and the absorbance

was read at 570 nm using an ELISA plate reader. Nitrite concentration was determined by comparison with reference to a sodium nitrite standard curve ([Temdie *et al.*, 2020](#)).

2.5.5.2|Evaluation of Levels of Hematological Parameters in Serum

This deals with the overall study of blood, which includes the causes, prognosis, treatment and prevention of diseases related to blood. Hematological analysis was performed using an automatic hematological analyzer (Sysmex KX-21N) according to the manufacturer's instructions. The parameters included: white blood cells (WBC), red blood cells (RBC),

Hemoglobin (HB), hematocrit, platelet, monocytes, granulocytes, neutrophils, eosinophils, basophils, Mean Corpuscular Hemoglobin (MCH) and Mean Corpuscular Hemoglobin Concentration (MCHC).

2.5.5.3|Evaluation of Inflammatory Cytokine Levels in Serum

Monoclonal antibodies specific for rats TNF- α , INF- γ , IL-1 β , IL-6 and IL-10 were pre-coated onto 96 well microplate (for each cytokine). 100 µL of standards and samples were pipetted into these wells. Using 400 µL of wash buffer, unbound substances were washed away and then incubation was done for 2 h at room temperature. An enzyme-linked polyclonal antibody specific for mice TNF- α , INF- γ , IL-1 β , IL-6 and IL-10 (100 µL) was added to the wells and incubated at room temperature for 2 h. Following a wash to remove any unbound antibody-enzyme reagent, a substrate solution (100 µL) was added to the wells. It was incubated for 30 min at room temperature in the dark. The enzyme reaction yielded to a blue product that turned yellow when the stop solution (50 µL) was added. The optical

density of each well was determined within 30 minutes, using an ELISA plate reader set to 450 nm. To obtain the standard curve, which was used to determine the concentration of the cytokines, serial dilutions of protein standards were made. Protein concentration was expressed as a function of the optical density (OD). The standard curves were used to calculate the concentration of TNF- α , INF- γ , IL-1 β , IL-6 and IL-10 respective. The intensity of the color measured was in proportion to the amount of TNF- α , INF- γ , IL-1 β , IL-6 and IL-10 bound in the initial step.

2.5.5.3| Histopathological Study of Organ Tissue

This was done according to the method of Temdie *et al.* (2012). The heart, kidney, liver and pancreas samples previously kept in formaline (10%) were subjected to the following histological techniques. After dehydration, the samples were embedded in paraffin to form blocks. 5 μ m thick sections from each sample were stained with haematoxylin-eosin and observed under the light microscope.

2.6| Statistical Analysis

Data was expressed as mean $\pm$ SD. Graphpad InStat version 3 was used for the statistical analysis to analyze data by one-way ANOVA with Bonferroni post-test for significance of differences between groups. P<0.05 were considered significant.

3| Results

3.1| Oil Quality Indices of Sardinella Maderensis Fish Oil

The physicochemical properties of *Sardinella maderensis* fish oil evaluated are presented in table 3. The free fatty acid value was evaluated in terms of acid value and converted to percentage as shown in the table with % free fatty acid value

with the least value of 0.98 which is slightly lower than the standard value.

Table 3: Characterization of Sardinella maderensis fish oil physicochemical properties

Parameter	Sardinella maderensis oil	WHO/FAO standard 2008
% Free fatty acid value	0.98 $\pm$ 0.27	1-2.5
Iodine value (gt ₂ /100g	118.42 $\pm$ 5.84	None
Peroxide value (meqO ₂ /Kg)	7.94 $\pm$ 0.39	<10
P-Anisidine value	18.57 $\pm$ 0.30	$\leq$ 20
TOTOX Value	34.24 $\pm$ 0.80	$\leq$ 40

Note: TOTOX: Total Oxidation. Fatty acids are expressed in term of percentage.

3.2| Animal Bioassay

3.2.1 |Relative Growth Response

Results for the relative growth response are presented on figure 1. All the groups except test group III had their highest relative growth response at week 2. The normal group, positive control and test group I recorded their lowest

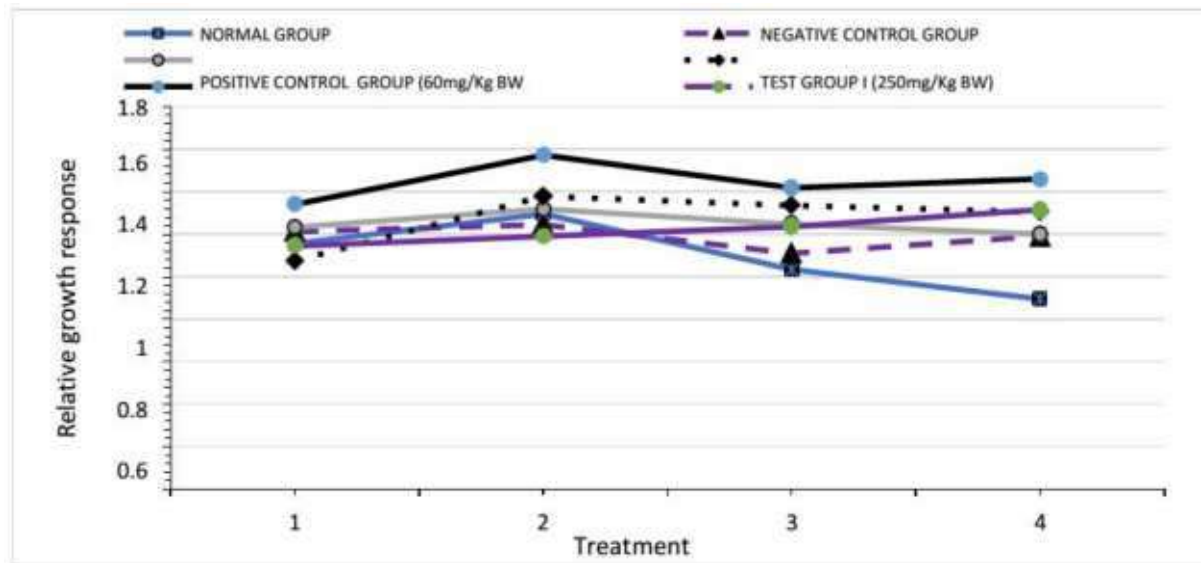


Figure 1: Relative growth response in rats during the period of treatment

growth response during the 4th week while the negative control group and test group II recorded their lowest at the 3rd week. Test group III increased from week 1 through to week 4.

3.2.2 | Organ Weights

The weights of the organs recorded immediately after sacrifice were statistically analyzed. Their mean $\pm$ SD are presented in Table 4.

According to statistical analysis, the brain, heart, liver, spleen, kidneys and lungs weights were not significantly different ($P > 0.05$) across all groups respectively. However, Pancreas weights for the negative control were significantly higher ($P < 0.05$) than that of the normal group while not significantly different as compared to the other groups. Also, there was no difference in pancreas weights amongst the test groups.

3.2.3 | Effect of *Sardinella maderensis* oil on oxidative stress biomarkers

3.2.3.1 | Effect of *Sardinella maderensis* oil on MDA levels

The results of MDA are presented on table 5. Comparing the negative control group to the test groups, brain MDA levels were not significantly different ($P > 0.05$) in the negative control group and test groups. Heart and liver MDA levels in the negative control group were significantly higher ($P < 0.05$) than test group III. MDA levels in the pancreas and serum were significantly higher ($P < 0.001$) in the negative control group in comparison to the test groups. Brain and liver MDA levels were not significantly different ($P > 0.05$) amongst the test groups while heart, pancreas and serum MDA levels in test group III had significantly lower ($P < 0.001$) MDA levels than test group I.

3.2.3.2 | Effect of *Sardinella maderensis* oil on GSH levels

Results of GSH concentrations are presented in table 6. Concentrations of GSH in the brain, heart and pancreas were not significantly different ($P > 0.05$) in the negative control group and test groups.

Table 4: Organ weights (g)

	Normal group	Negative control	Positive control	Test group I	Test group II	Test group III
Brain (g)	1.62±0.13	1.61±0.16	1.65±0.14	1.6±0.11	1.71±0.19	1.78±0.14
Pancreas(g)	0.29±0.14 ^a	0.96±0.21 ^b	1.03±0.28 ^{cb}	0.8±0.46 ^{abcd}	0.52±0.27 ^{abcde}	0.64±0.40 ^{abcde}
Heart (g)	0.74±0.07	0.75±0.07	0.79±0.09	0.85±0.09	0.8±0.15	0.84±0.10
Liver (g)	8.13±1.81	7.65±1.01	7.1±0.89	7.36±0.56	8.17±1.95	8.76±0.46
Spleen(g)	0.7±0.14	0.64±0.13	0.64±0.23	0.71±0.12	0.85±0.33	0.84±0.17
Lungs(g)	1.55±0.14	1.71±0.36	1.38±0.33	1.5±0.06	1.53±0.33	1.52±0.16

Note: groups with same letters (a, b, c, d, e) for the pancreas are not significantly different. For example, normal group for the pancreas is not significantly different from all three test groups.

The negative control group had a significantly lower ($P<0.01$) liver GSH concentration as compared to test group II and III but, the difference was not significant ($P > 0.05$) as compared to test group I.

Table 5: Effect of *Sardinella maderensis* oil on MDA levels in organ homogenate and serum

MDA levels (μmol/g)						
	Normal group	Negative control	Positive control	Test group I	Test group II	Test group III
Brain	2.67±0.06 ^a	10.33±1.25 ^b	4.05±1.44 ^{abc}	7.96±0.98 ^{abcd}	7.62±3.98 ^{abcde}	6.02±1.02 ^{abcde}
Heart	12.68±0.0 ^a	34.69±6.18 ^b	16.92±5.02 ^{ac}	34.50±3.46 ^{chf}	24.47±0.0 ^{hulefg}	20.01±0.52 ^{aeig}
Pancreas	1.31±0.02 ^a	23.04±0.17 ^b	19.68±2.5 ^{bce}	15.70±0.52 ^{de}	4.57±2.70 ^{af}	3.50±1.37 ^{af}
Liver	0.02±0.01 ^a	1.15±0.33 ^b	0.47±0.21 ^{abc}	0.44±0.19 ^{abcd}	0.59±0.02 ^{abcde}	0.43±0.23 ^{acde}
Serum (μmol/l)	1.73±0.22 ^a	7.42±0.02 ^b	4.54±0.00 ^{cc}	4.37±0.00 ^{de}	2.67±0.90 ^{af}	2.45±0.17 ^{af}

Note: MDA: Malondialdehyde. Groups with same letter (n=5, a, b, c, d, e, f and g) are not significantly different ($P>0.05$). For example, brain, normal group (a) is only significantly different to the negative control group (without a)

Serum GSH levels in the negative control group were significantly different ($P<0.001$) from test group II and III. Amongst the test groups of

organ homogenates, GSH levels were not significantly different ($P>0.05$). Serum GSH levels were significantly higher ($P<0.001$) in test group III than in test group I and II.

Table 6: Effect of *Sardinella maderensis* oil on GSH levels in organ homogenate and serum

GSH levels (μmol/g)						
	Normal group	Negative control	Positive control	Test group I	Test group II	Test group III
Brain	6.36±0.10 ^a	2.63±1.74 ^b	2.61±0.41 ^{bc}	2.83±0.83 ^{bed}	2.93±0.01 ^{bcde}	3.36±0.29 ^{bcdef}
Heart	12.13±0.21 ^a	7.36±0.36 ^b	7.37±1.17 ^{bc}	7.81±1.64 ^{bed}	7.90±0.47 ^{bcde}	8.38±0.16 ^{bcdef}
Pancreas	17.47±0.10 ^a	3.47±1.01 ^b	6.47±0.00 ^{bc}	7.61±0.00 ^{bed}	6.60±1.69 ^{bcde}	6.97±4.76 ^{bcdef}
Liver	9.58±1.64 ^a	4.81±0.27 ^b	5.88±0.62 ^{bc}	5.88±1.82 ^{bed}	9.49±0.00 ^{ade}	8.35±0.00 ^{ade}
Serum	79.63±0.47 ^a	24.93±0.73 ^b	66.10±0.14 ^c	30.97±9.75 ^{bd}	48.62±5.15 ^e	63.82±7.07 ^{cf}

Note: GSH: Reduced glutathione. Groups with same letters (n=5; a, b, c, d, e and f) are non-significantly different ($P>0.05$). For example, brain negative control (b) is not significantly different from all groups except the normal group (without b)

3.2.3.3| Effect of *Sardinella maderensis* oil on SOD activity

The results of SOD activity are presented on table 7. There was no significant difference ($P>0.05$) amongst all the groups for organ homogenates and serum.

Table 7: Effect of *Sardinella maderensis* oil on SOD activity in organ homogenate and serum

SOD activity(U/mg)						
	Normal group	Negative control	Positive control	Test group I	Test group II	Test Group III
Brain	1.27±0.01	1.33±0.02	1.25±0.02	1.26±0.02	1.40±0.34	1.11±0.03
Heart	3.15±0.23	2.45±0.30	3.04±0.06	2.52±0.72	2.65±0.29	2.91±0.09
Pancreas	1.77±0.21	1.39±2.80	2.60±0.46	2.01±0.00	2.08±0.00	1.93±0.00
Liver	0.23±0.00	0.30±0.04	0.28±0.03	0.26±0.00	0.25±0.05	0.25±0.09
Serum (U/L)	40.22±0.37	39.8±0.00	39.8±0.00	40.04±0.00	39.83±0.12	39.87±0.05

Note: Expressed as SOD unit/mg of tissue. Where SOD: Superoxide Dismutase. No letters signifying that there is no significant difference amongst the groups of each organ and serum.

Table 8: Effect of *Sardinella maderensis* oils on CAT activity in organ homogenate and serum

	CAT activity (U/mg)					
	Normal group	Negative control	Positive control	Test group I	Test group II	Test group III
Brain	14.15±0.81 ^a	0.48±0.00 ^b	1.65±0.00 ^{cb}	2.32±0.00 ^{cd}	3.37±0.00 ^{de}	4.48±0.66 ^{ef}
Heart	49.70±10.75 ^a	12.76±4.20 ^b	22.71±0.00 ^{bc}	18.90±0.10 ^{bcd}	18.88±0.00 ^{bce}	27.03±3.52 ^{bcf}
Pancreas	110.79±4.91 ^a	14.63±1.52 ^b	54.54±21.21 ^c	14.19±6.76 ^{bd}	15.84±11.73 ^{bde}	25.06±0.01 ^{bcdff}
Liver	81.72±19.23 ^a	11.63±4.74 ^b	43.20±0.24 ^{ac}	22.13±6.73 ^{bceff}	26.63±5.16 ^{bdefg}	58.30±14.47 ^{acfig}
Serum	74.15±0.00 ^a	22.02±0.00 ^b	51.14±5.85 ^{bc}	30.18±12.46 ^{bd}	40.71±2.92 ^{bc}	61.89±8.45 ^a

Note: Expressed as μ moles of H₂O₂ decomposed/min/mg (U/mg) of tissue. Groups with the same letter (n=5, a, b, c, d, e, f and g) are not significantly different (P>0.05). For example, brain CAT levels, the negative control group (b) is not significantly different only to the positive control group

3.2.3.4|Effect of *Sardinella maderensis* oil on CAT activity

The results of CAT activity are presented on table 8. Brain CAT activity level was significantly lower (P<0.001) in the negative control group than in the test groups. Heart and pancreas CAT activities of the negative control group were not significantly different compared to test groups. Liver and serum CAT levels were significantly lower (P<0.01) in the negative control group than in test group III but, no significantly different (P>0.05) as compared to the test group I and II. Comparison amongst the test groups revealed that brain and serum CAT activities in test group III was significantly higher (P<0.01) as compared to test group I and not-significant as compared to test group II (P>0.05). Levels of CAT activity in the liver and pancreas amongst the test groups were not significantly different (P>0.05).

3.2.3.5| Effect of *Sardinella maderensis* oil on Nitric levels

The results of Nitrite levels are presented in table 9. Brain and serum nitrite levels were not significantly different in the negative control group as compared to the test groups. Heart nitrite levels in the negative control group was significantly higher (P<0.001) than in test groups. Pancreas and liver nitrite levels were significantly higher (P<0.01) in the negative control group than in test groups II and III but, not significantly different (P>0.05) as compared test group I. Brain, liver and pancreas nitrite levels amongst the test groups were similar. Meanwhile, heart and serum in test group III had significantly (P<0.01) lower nitrite levels as compared to test group I.

Table 9: Effect of *Sardinella maderensis* oil on Nitrite levels in organ homogenate and serum

	Nitrite levels ($\mu\text{mol/g}$)			
	Normal group	Negative control	Positive control	Test group
Brain	5.91 $\pm$ 0.82 ^a	9.01 $\pm$ 1.37 ^b	7.34 $\pm$ 0.00 ^{abc}	Test group I
				7.34 $\pm$ 0.77 ^{abcde}
Heart	7.73 $\pm$ 0.55 ^a	25.17 $\pm$ 0.00 ^b	8.08 $\pm$ 0.16 ^{ac}	Test group II
				13.23 $\pm$ 2.96 ^{defg}
Pancreas	15.50 $\pm$ 0.69 ^a	102.88 $\pm$ 36.6 ^b	56.72 $\pm$ 0.0 ^{abd}	Test group III
				6.26 $\pm$ 0.22 ^{abcde}
Liver	8.58 $\pm$ 2.30 ^a	26.72 $\pm$ 0.00 ^b	12.33 $\pm$ 4.46 ^{ac}	8.35 $\pm$ 1.75 ^{acg}
				24.20 $\pm$ 0.1 ^{acdef}
Serum	12.50 $\pm$ 0.27 ^a	19.90 $\pm$ 4.82 ^{ac}	17.03 $\pm$ 0.0 ^{acd}	11.11 $\pm$ 3.88 ^{acde}
				12.22 $\pm$ 1.58 ^{acde}

Groups carrying different letters (n=5, a, b, c, d, and e) are significantly different. For example, brain nitrite levels; the normal group (a) is only significantly different from the negative control group (b).

Table 10: Effects of *Sardinella maderensis* oil on hematological parameters

	Normal group	Negative control	Positive control	Test group I	Test group II	Test group III
WBC($\times 10^3/\mu\text{L}^{-1}$)	11.4 $\pm$ 0.03 ^a	8.62 $\pm$ 0.71 ^b	11.13 $\pm$ 0.02 ^a	12.22 $\pm$ 0.10 ^{ac}	13.12 $\pm$ 0.84 ^{cl}	12.5 $\pm$ 0.01 ^{adl}
RBC($\times 10^6/\mu\text{L}^{-1}$)	6.32 $\pm$ 0.30 ^a	4.60 $\pm$ 0.67 ^b	6.16 $\pm$ 0.09 ^{ac}	6.55 $\pm$ 0.47 ^{acd}	6.32 $\pm$ 0.25 ^{acde}	6.58 $\pm$ 0.06 ^{acde}
Hb (g/dL)	11.9 $\pm$ 0.10 ^a	8.75 $\pm$ 0.79 ^b	11.09 $\pm$ 0.90 ^{ac}	11.88 $\pm$ 0.33 ^{acd}	12.61 $\pm$ 0.02 ^{ade}	12.71 $\pm$ 0.78 ^{ade}
Hematocrit(%)	36.45 $\pm$ 0.30 ^a	29.83 $\pm$ 0.81 ^b	35.57 $\pm$ 0.14 ^{ac}	35.58 $\pm$ 0.01 ^{acd}	36.15 $\pm$ 1.06 ^{acde}	36.09 $\pm$ 0.04 ^{acde}
Platelets($\times 10^3/\mu\text{L}^{-1}$)	455 $\pm$ 5.66 ^a	421 $\pm$ 2.83 ^b	458 $\pm$ 8.49 ^{ac}	448 $\pm$ 1.41 ^{acd}	455 $\pm$ 2.83 ^{ade}	454 $\pm$ 18.38 ^{acde}
Lymphocytes	78.42 $\pm$ 0.03 ^a	61.05 $\pm$ 5.14 ^b	78.79 $\pm$ 0.13 ^{ac}	76.94 $\pm$ 0.89 ^{acd}	78.08 $\pm$ 0.54 ^{acde}	78.39 $\pm$ 1.78 ^{acde}
Monocytes (%)	6.9 $\pm$ 0.01 ^a	4.82 $\pm$ 0.85 ^b	6.68 $\pm$ 0.13 ^{ac}	6.51 $\pm$ 0.57 ^{acd}	6.92 $\pm$ 0.01 ^{ade}	6.89 $\pm$ 0.08 ^{acde}
Granulocytes (%)	20.8 $\pm$ 0.08 ^a	16.83 $\pm$ 0.74 ^b	19.7 $\pm$ 0.19 ^c	20.43 $\pm$ 0.14 ^{ad}	20.33 $\pm$ 0.02 ^{ade}	20.65 $\pm$ 0.28 ^{ade}
Neutrophils (%)	11.29 $\pm$ 0.06 ^a	9.05 $\pm$ 0.18 ^b	9.55 $\pm$ 0.31 ^c	10.72 $\pm$ 0.23 ^d	11.54 $\pm$ 0.25 ^a	11.14 $\pm$ 0.07 ^h
Eosinophils (%)	2.86 $\pm$ 0.14 ^a	1.56 $\pm$ 0.18 ^b	2.31 $\pm$ 0.06 ^c	2.82 $\pm$ 0.08 ^d	2.88 $\pm$ 0.06 ^{ade}	2.94 $\pm$ 0.07 ^{ade}
Basophils (%)	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
MCV (fL)	55.93 $\pm$ 0.85 ^a	52.07 $\pm$ 0.63 ^b	55.4 $\pm$ 0.19 ^{ac}	55.85 $\pm$ 0.82 ^{acd}	56.03 $\pm$ 0.71 ^{acde}	56.03 $\pm$ 0.71 ^{acde}
MCH (pg)	17.71 $\pm$ 0.06 ^a	14.53 $\pm$ 0.13 ^b	16.4 $\pm$ 0.19 ^{ce}	16.35 $\pm$ 0.02 ^{de}	17.46 $\pm$ 0.25 ^{af}	17.71 $\pm$ 0.35 ^{af}
MCHC (g/dL)	29.5 $\pm$ 0.65 ^a	28.03 $\pm$ 0.69 ^b	28.48 $\pm$ 0.07 ^{abd}	29.36 $\pm$ 0.13 ^{ade}	30.03 $\pm$ 0.08 ^{ac}	28.95 $\pm$ 0.70 ^{ade}

Note: WBC; White Blood Cells; RBC; Red Blood Cells, Hb; Hemoglobin, MCV; Mean corpuscular volume, MCH, Mean corpuscular hemoglobin, MCHC; Mean corpuscular hemoglobin concentration. Groups with same letter (n=6, a, b, c, d, e and f) are not significantly different ($P>0.05$).

3.2.4| Effect of *Sardinella maderensis* oil on hematological parameters

The results of Hematological parameter evaluation are presented on table 10. The result showed significantly lower ($P<0.001$) concentrations of WBC, HB, hematocrit,

platelets, lymphocytes, monocytes, granulocytes, neutrophils, eosinophils, basophils, MCH and MCHC in the negative control group as compared to the other groups. Test group III of the WBC was not significantly different ($P>0.05$) from test group I and II. On the contrary, RBC, Hb, hematocrit, platelets, lymphocytes, monocytes, granulocytes, eosinophils, basophils revealed no significant difference amongst the test groups. Neutrophils showed significant differences amongst the test groups where, test group III was significantly higher ($P<0.05$) than test group I and II. MCH had a significant difference amongst its test groups, where, test group III was significantly higher ($P<0.001$) than test group I only.

3.2.5 | Effect of *Sardinella maderensis* fish oil on inflammatory cytokine levels

Results for inflammatory cytokine are presented on table 11. Serum levels of TNF- α , INF- δ and IL-1 β were significantly higher in the negative control group as compared to the normal group, positive control, test group II and III ($P<0.001$), whereas, not significantly different as compared to the test group I ($P>0.05$). Their concentrations in test group III were significantly lower ($P<0.001$) than in the test group I and II. Serum levels of IL-6 and IL-10 were significantly higher ($P<0.001$) in the negative control group as compared to the other groups. Their concentrations in test group III were significantly lower ($P<0.001$) as compared to the test group I and non-significant as compared to test group II ($P>0.05$).

3.2.6 | Effects of *Sardinella maderensis* oil on histology

Heart histology

The heart histology performed (Figure 2) revealed a muscle fiber degeneration in the tissue of the negative control group which is also

revealed to be restored in the *Sardinella maderensis* oil treated groups.

Table 11: Serum levels of inflammatory cytokines

	Normal Group	Negative control	Positive control	Test group I	Test group II	Test group III
TNF- α (pg/ml)	172.12 $\pm$ 19.93 ^a	374.4 $\pm$ 16.69 ^b	180.1 $\pm$ 8.9 ^{ac}	354 $\pm$ 31.26 ^{cb}	246 $\pm$ 12.88 ^d	189.88 $\pm$ 7.84 ^{ac}
INF- δ (pg/ml)	137.18 $\pm$ 15.02 ^a	380.91 $\pm$ 8.22 ^b	162.8 $\pm$ 20.79 ^{ac}	357 $\pm$ 35.26 ^{cb}	204.3 $\pm$ 17.04 ^d	168.05 $\pm$ 5.07 ^{ac}
IL-1 β (pg/ml)	144.78 $\pm$ 8.46 ^a	334.45 $\pm$ 23.03 ^b	154.1 $\pm$ 11.75 ^{ac}	315.81 $\pm$ 21.95 ^{cb}	204.8 $\pm$ 31.74 ^d	167.68 $\pm$ 14.32 ^{ac}
IL-6 (pg/ml)	307.4 $\pm$ 4.58 ^a	467.33 $\pm$ 13.36 ^b	304.6 $\pm$ 11.91 ^{ad}	412.28 $\pm$ 37.14 ^e	333.2 $\pm$ 17.05 ^{ade}	309.03 $\pm$ 13.41 ^{ade}
IL-10 (pg/ml)	410.34 $\pm$ 22.6 ^b	205.94 $\pm$ 8.87 ^a	222.6 $\pm$ 7.31 ^{ac}	355.58 $\pm$ 15.95 ^c	242.8 $\pm$ 28.03 ^{def}	234.36 $\pm$ 12.09 ^{acdf}

Note: TNF- α ; Tumor necrosis factor- α , INF- δ ; Interferon- δ , IL-1 β ; Interleukin-1 β , IL-6; Interleukin-6, IL-10; Interleukin-10. Groups with same letter (n=5, a, b, c, d and e) are non-significantly different ($P>0.05$)

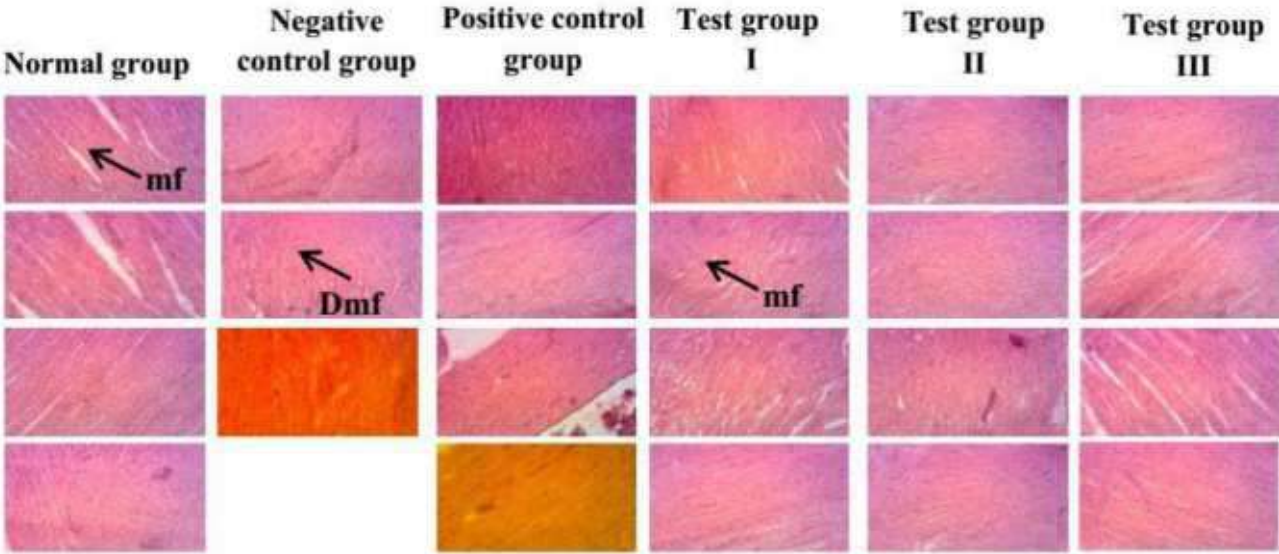


Figure 2: Heart tissue histology. Mf=Muscular fiber, DMf=Degeneration of Muscular fiber

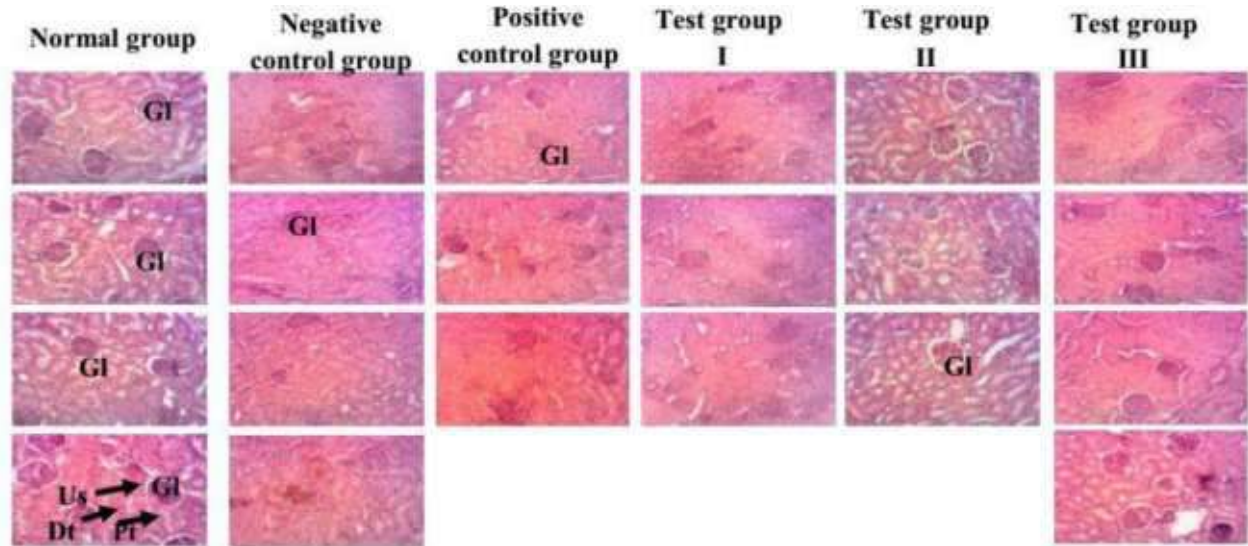


Figure 3: kidney tissue histology. Gl = Glomerulus, Us = Urinary space,Dt = Distal tubule, Pt = Proximal tubule, NUs = Narrowing of Urinary space

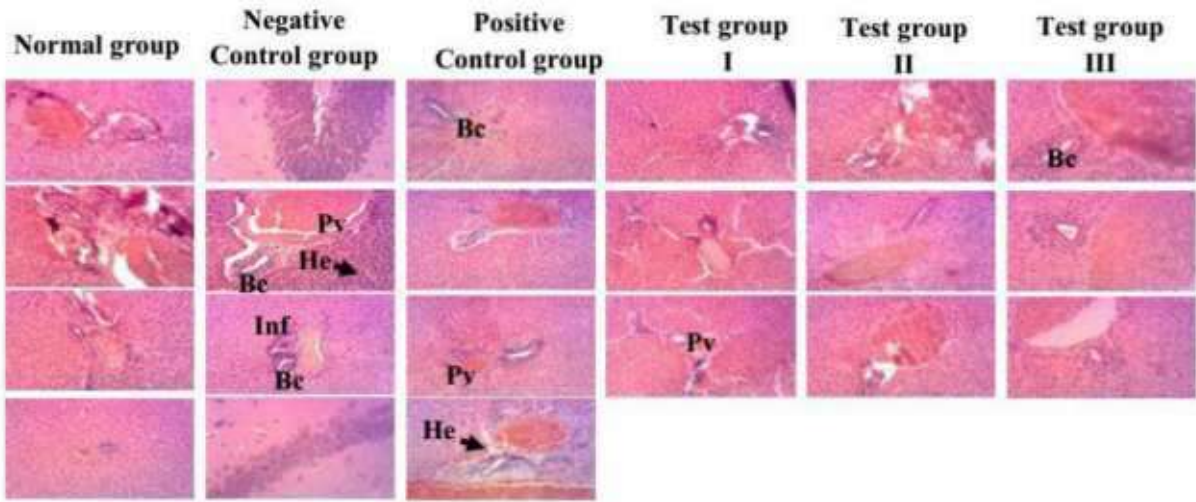


Figure 4: liver tissue histology. Pv = Portal vein, He = Hepatocyte, Sc =Sinusoid capillary, Bc = Biliary canalicus, Inf= Leukocytes infiltration

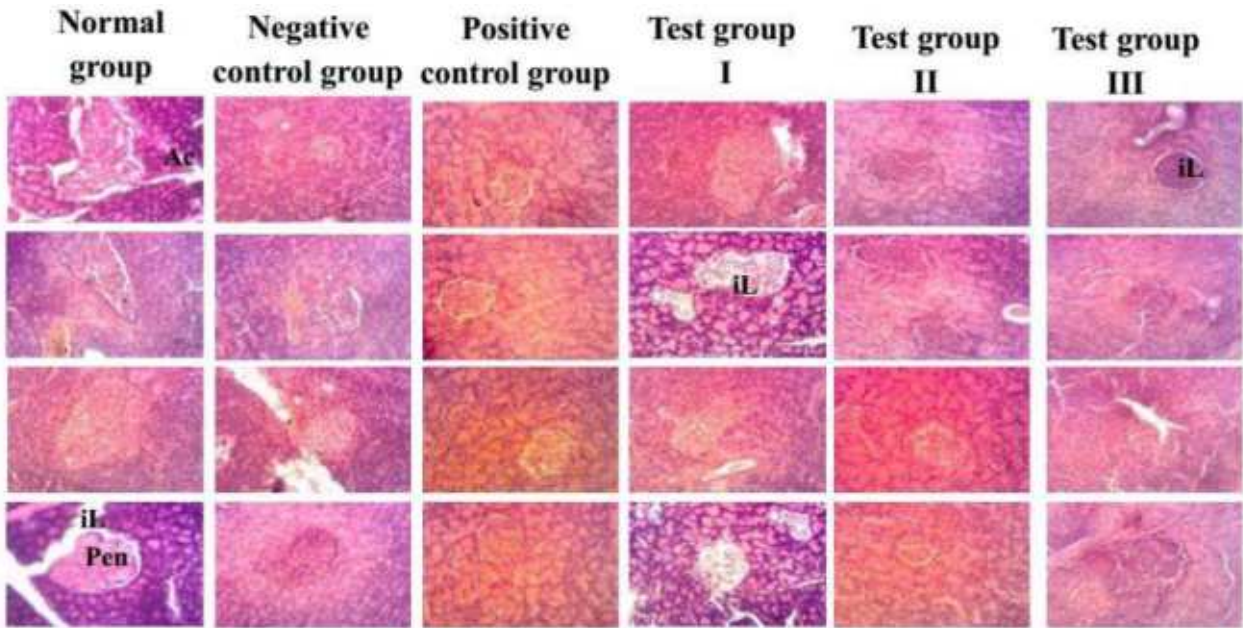


Figure 5: Pancreas tissue histology. IL = islets of Langerhans, Ac = Acini, Pen = endocrine pancreas, Pex = exocrine pancreas.

Kidney histology

Histological study of the kidney presented on figure 3 revealed that the urinary space was

narrowed in the negative control group. The tissue section isn't clearly seen in the negative control as compared to the test group II and III.

Liver histology

The histopathology of the liver is presented on figure 4. The liver section of Negative control shows a diffuse infiltration of leukocytes in the parenchyma. Some leukocyte infiltrations loci were still observed in the parenchyma of the group II.

Pancreas histology

Histological study of the pancreas section is shown in figure 5. In the negative control group, a significant reduction in size of the islets of Langerhans was observed compared to normal control. The test group III show some disorganization of the whole structure of the islets. The normal group showed the exocrine section with well-arranged acinar cells.

4 | Discussion

4.1 | Physicochemical Properties of *Sardinella maderensis* oil

The % free fatty acid value (0.98) recorded in this study is within the range ($\leq 2.5\%$) recommended by the [Codex Alimentarius \(2017\)](#) and this is in accordance with a % free fatty acid value of 0.91% reported by [Adepoju *et al.* \(2014\)](#) for soursop seed oil (which is also of good nutritional quality) but way lower than the results obtained by [Tenyang *et al.* \(2020\)](#) who worked on fish oil obtained from far north of Cameroon. This indicates the presence of low free fatty acids in the oil. FFAs are known to negatively affect protein solubility and also cause texture deterioration ([Tenyang *et al.*, 2020](#)). Accumulation of FFAs in fish oil is undesirable, since it may lead to catalyzed reaction such as increased susceptibility to oxidation and development of off-flavor. Percentage free fatty acid values in this study indicate that the oil samples can be stored for long.

The iodine value shows the degree of unsaturation of oil. The Value recorded in this study for *Sardinella maderensis* oil (118.42 gI₂/100g) was higher than 95.27 gI₂/100g for raw mullet obtained by [Tenyang *et al.* \(2020\)](#) and 148.73 gI₂ /100g for *Sardinella maderensis* oil obtained by [Manz *et al.* \(2023\)](#) but, lower than the value obtained by [Manz *et al.* \(2021\)](#) for *Sardinella maderensis* oil (180.45 gI₂/100g). The changes in values between fish oils may be as a result of their environment, diet or fishing period ([Mourete *et al.*, 2006](#)). Generally, during lipid oxidation, the double bonds of unsaturated fatty acids are attacked by free radicals, resulting in the reduction of the number of unsaturation ([Tynek *et al.*, 2001](#)). A relatively high iodine value reflects a high content of unsaturated fatty acids ([Tan and Chow, 2010](#)).

The peroxide value in this study (7.94 meq/kg) is indicative of low deterioration for the *Sardinella maderensis* oil. The peroxide value is a measure of primary oxidation products present in fat sample. Rancidity increases peroxide values and decreases the quality of the fish oil. That is, Oxidation of the fish oil makes rancidity and off-flavors higher ([Sugeng *et al.*, 2016](#)). The peroxide value obtained is lower than 11.93 meq/kg obtained by [Sugeng *et al.* \(2016\)](#) in crude sardine oil. The rate of degradation of peroxides to other compounds and its reaction with other substances may exceed the generation of new ones and this may result in a low peroxide value ([Raharjo, 2008](#)).

p-anisidine test is used as a measure of the level of secondary oxidation products. [Guillen & Cabo \(2002\)](#) stated that the high peroxide value can cause high p-anisidine values if the process given allows fish oil to undergo further degradation. The TOTOX value obtained by [Sugeng *et al.* \(2016\)](#) in crude sardine oil (35.47 meq/kg) is

slightly higher than the value obtained in this study (34.24 meq/kg). The TOTOX value obtained is in the range of standard (≤ 40) recommended by [Codex Alimentarius Commission \(2021\)](#).

4.2| Relative Growth Response

Specific markers for both mice and rats that define the presence or absence of obesity are absent. It is considered by a majority of studies that the differences in total body weight gain can be used as the main parameter to assess the outcome of the development of obesity ([De-Moura *et al.*, 2021](#)). In this study, the general increase in growth response at the second week can be due to increase in the rat's lean mass, bone mass and fat mass, not necessarily due to obesity. In an attempt to explain the various effects of the fish oil on the relative growth response as observed in this study, we report that a decrease in the relative growth response for test groups I and II could depict that administration of *Sardinella maderensis* oil initiated an action whereby the anabolic effects synergized with the catabolic effects in protecting against excessive body fat deposition. The intake of fish oil rich in omega-3 fatty acids produced slight reductions in body weight. The observed reduction in the relative growth response in this present study suggests that omega-3 fatty acids does not boost adiposity or enhance body fat accumulation but rather modulates fat accumulation in tissues by enhancing adipocyte apoptosis ([Nnamonu *et al.*, 2020](#)). Perhaps omega-3 PUFAs were more effective in preventing weight gain rather than assisting in weight loss, causing loss of appetite and reduction of fat ([Munro & Garg 2013](#)).

4.3| Organ Weight

The organ weight to body weight ratio gives a proportional size of the organ to body weight. It

has been suggested that the use of organ-body weight ratios may be valuable in evaluating the relationship between certain experimental situations and the biological response of a test organism. [Piao *et al.* \(2013\)](#) said organ's weight can be the most sensitive indicator of an effect of an experimental compound, as significant differences in organ weight between treated and untreated (control) animals show the absence or presence of any morphological changes.

4.4| Oxidative Stress

Research shows that a diet high in fat and carbohydrates induces a significant increase in oxidative stress and inflammation in persons with obesity ([Patel *et al.*, 2007](#)). ROS have been found to be involved in the control of body weight by exerting different effects on hypothalamic neurons, which control satiety and hunger behavior ([Horvath *et al.*, 2009](#)). Oxidative stress biomarker, MDA are the products of the peroxidation of PUFAs ([Fernández- Sánchez *et al.*, 2011](#)). In this study, obesity-elevated lipid peroxidation examined by serum levels of TBARS was significantly reversed by *Sardinella maderensis* oil and this was in accordance to work done by [Liu *et al.* \(2021\)](#). In accordance to work done by [Vincent *et al.* \(2004\)](#), total antioxidant status (TAS) in the negative control group of this study decreased whereas high levels were observed in normal group. In relation to several diseases, serum TAS is consistently decreased ([Kwak & Yoon, 2007](#)). In contrast to this study, negative associations have been reported between nitrite/nitrate levels and obesity ([Piva *et al.*, 2013](#)). [Ueyama *et al.* \(2008\)](#) concluded that increased nitrite levels are associated with a clustering of MetS components.

Brain CAT activity level was significantly lower in the negative control group than in the test groups. Heart and pancreas CAT activities of the negative

control group were not significantly different compared to test groups. This may be mediated by the presence of bioactive compounds in *Sardinella maderensis* oil, such as phenolic compounds and flavonoids, which according to [Cadeira *et al.* \(2022\)](#) can activate signaling pathways involved in the regulation of CAT gene expression. *Sardinella maderensis* oil contains high levels of antioxidants, which can scavenge free radicals.

GSH is the most abundant antioxidant molecule in cells, and its depletion is a hallmark of oxidative stress ([Lui *et al.*, 2022](#)). Therefore, measurement of GSH levels is often used as an indicator of oxidative stress in various biological systems. Concentrations of GSH in the brain, heart and pancreas were not significantly different in the negative control group and test groups. The negative control group had a significantly lower liver GSH concentration as compared to test group II and III but, the difference was not significant as compared to test group I. Serum GSH levels in the negative control group were significantly different from test group II and III. This could be attributed to the high antioxidant capacity of *Sardinella maderensis* oil. This results are similar to that of [Manzkoule *et al.* \(2023\)](#).

Heart nitrite levels in the negative control group was significantly higher than in test groups. This could be attributed to the the high concentration of omega-3 polyunsaturated fatty acids Pancreas and liver nitrite levels were significantly higher in the negative control group than in test groups II and III but, not significantly different as compared test group I ([Bhushan *et al.*, 2014](#)).

It was observed that the was no significant difference amongst all the groups for organ homogenates and serum *Sardinella maderensis* significantly increases the levels and activity of key

enzymes like SOD. Thus increasing the antioxidant defenses ([Koule *et al.*, 2023](#)).

4.5| Hematological Parameters

Hematological parameters are widely used clinical indicators of health and disease. The increase in the levels of hematological parameters in the tests groups as recorded on table 11 can be explained by the fact that, fish-oil-rich diet promotes hematopoiesis in the bone marrow and spleen ([Xia *et al.*, 2015](#)). In this study, there was a decrease in white blood cell count and its subtypes following obesity induction. According to [Trottier *et al.* \(2012\)](#), HFD induced obesity significantly altered hematopoietic and lymphopoietic functions in the bone marrow and thymus. Eating a high fat diet may compromise the development of hematopoietic stem and progenitor cells (HSPCs) ([Mincham *et al.*, 2021](#)). The platelet counts were significantly decreased in the negative group in accordance to [Harishankar *et al.* \(2011\)](#). Also, in accordance to [Haidari *et al.* \(2020\)](#), hemoglobin and hematocrit levels were significantly lower in obese rates. Nonetheless pre-treatment with *Sardinella maderensis* oil significantly increased white blood cell count and its subtypes. Comparing the treatment groups and the negative control group, there was a significant increase in the white blood cell count and its subtypes including monocytes, granulocytes, and neutrophils. All three doses of *Sardinella maderensis* oil, that is, the test groups significantly increased the levels of the hematological parameters like white blood cells (WBC), red blood cells (RBC), Hemoglobin (HB), hematocrit, platelet, monocytes, granulocytes, neutrophils, eosinophils, basophils, Mean Corpuscular Hemoglobin (MCH) and Mean Corpuscular Hemoglobin Concentration (MCHC) and this results are in accordance with [Dizaye & Jarjees \(2014\)](#).

4.6| Inflammatory Cytokines

Clearing the leukocytes from lesions, resolving and restoring tissue homeostasis is the ultimate target of any inflammatory process (Van Dyke and Kornman, 2008). According to Kim *et al.* (2008), obesity elevates the number of blood cytokines such as TNF- α , IL-1 β , IL-8 and IL-6 in the circulation. TNF- α are the most evidently increased in the serum and white adipose tissue of obese individuals (Hotamisligil *et al.*, 1993; Farhangi *et al.*, 2013) and their levels are also noted to decrease with weight loss (Bastard *et al.*, 2000; Kern *et al.*, 1995). TNF- α , IL-6 and IL-1 are the most well-known mediators of the early inflammatory response (Margioris, 2009). The production of pro-inflammatory cytokines can be reduced by fish oil (due to the presence of EPA) (Calder, 2006; Morita, 2002). According to Calder (2010) and Ramirez-Ramirez *et al.* (2013) fish oil supplementation is effective in reducing the levels of proinflammatory cytokines. EPA and DHA give rise to resolvins and related compounds (e.g., protectins) through pathways involving cyclooxygenase and lipoxygenase enzymes (Serhan *et al.*, 2000; Serhan *et al.*, 2002; Serhan *et al.*, 2008). According to Calder (2010), in animal feeding studies and in cell culture, these mediators have been demonstrated to be anti-Inflammatory and inflammation resolving. Resolvin D1 inhibits IL-1 β production and protectin D1 inhibits TNF- α and IL-1 β production (Serhan *et al.*, 2008). This work was in accordance with a similar study carried out by Schmidt *et al.* (2015) where there were significant elevations of IFN- γ and TNF- α in cases of obesity.

4.7| Histological Study of Organs

In this study, the histopathological study of the pancreas of the negative control group showed a significant reduction in the islets of Langerhans size as compared to the normal control group.

Obesity promotes inflammation; persistent low-grade inflammation creates an environment that facilitates the induction and progression of pancreatic diseases including acinar cell necrosis (Gukovsky *et al.*, 2013). The heart histology of the normal group showed muscular fiber containing several nuclei. In negative group, leukocyte degeneration of muscular fiber resulting decreased in number of nuclei. According to studies carried out by Tansey *et al.* (2005), pericardial fat in cases of obesity not only surrounds, but can also penetrate into the wall of the ventricle and replace the myocardial muscle tissue. All the treatment group restored the architecture of the cardiac tissue. In normal group, a normal liver parenchyma in which a centrilobular vein, biliary canaliculus, hepatocytes separated by sinusoidal capillaries are well identified. The liver section of Negative group presents a hepatocyte swelling which is in accordance with Harishankar *et al.* (2011) and a diffuse infiltration of leukocytes in the parenchyma. The liver of animals in the test groups showed architecture quite similar to that of the normal group where the most parts of the tissue were well identified. This is an indication that the negative changes observed in the negative control group were attenuated by the administration of fish oil and this is in accordance to the study performed by Hwang *et al.* (2022). A diet high in fish oil induces a low degree of liver inflammation (Lionetti *et al.*, 2014). However, some leukocyte infiltrations loci were still observed in the parenchyma of test group II. The kidney section of the normal rat showed a parenchyma in which the glomeruli, Bowman's space (or Urinary space), proximal and distal tubes are well differentiated. However, the negative control group presents major disorganization in the parenchyma where proximal and distal tubules are not well observed,

the Bowman space strongly reduced. The treatment with fish oil at all the doses showed an improvement in kidney with a quite normal parenchyma in which glomeruli and Bowman space, distal and proximal tubes appear well identified.

5| Conclusion

Heart MDA levels in the negative control group (34.69 μ mol/g) were only significantly higher than test group III, 20.01 μ mol/g. Liver GSH levels in the negative control group (4.81 μ mol/g) were significantly lower than test group III, 8.35 μ mol/g. Catalase activity in the negative control group of the heart, 12.76, were significantly lower than in test groups I, II and III. Hematological parameters were all significantly lower in the negative control group than the test groups. Serum levels of pro-inflammatory cytokines in the negative control group, TNF- α (374.4), INF- δ (380.91) and IL-1 β (334.45) were significantly higher than in test group II and III. Histopathological studies in the negative control group showed a reduction in size of the islets of Langerhans, heart muscle fiber degeneration, hepatocyte swelling and the Bowman space strongly reduced. All of which were restored in the test groups administered fish oil. Rats administered fish oil had consistently lower values of oxidative stress biomarkers as compared to the negative control group showing that fish oil had a positive effect in reducing oxidative stress levels. Hence, *Sardinella maderensis* oil have a role in protection from the effects of obesity on oxidative stress, hematology, inflammatory cytokines and histology. Also, fish oil treated groups had improved levels of the hematological parameters, decreased levels of inflammatory cytokines and improved organ histology results. Hence, it can be concluded that *Sardinella maderensis* oil, rich in omega-3 PUFAs have a crucial role in curing and

in the management of obesity and its effects on oxidative stress, hematological, inflammatory cytokines and histological parameters.

|Acknowledgments

Not applicable.

|Conflict of interest

The authors declare that they have no conflict of interest.

|Data availability statement

The data used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

|Ethical approval

An ethical clearance for animal handling and care was obtained from the University of Buea-Institutional Animal Care and Use Committee, UB- IACUC N° 08/2023.

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